YASH DANIEL INGLE

Embedded Systems Engineer – C/C++, Firmware & Linux, RTOS, ARM/STM32, Sensor Interfaces (I^2C/SPI/UART), Debug/CI

Irvine, CA | +1 (949)524-7009 | yashingle1207@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Master of Embedded and Cyber-Physical Systems - University of California, Irvine, CA — ***GPA: 4.0 -*** (December 2025)

(Embedded Systems Modelling, Control Systems, Wireless Sensor Networks, IoT Systems, I/O Interface Programming, Embedded Linux/Yocto)

B. Tech - Computer Science with AI honors - MGM`s Jawaharlal Nehru Engineering College, India — ***CGPA: 9.16/10 -*** (July 2024)

(Data Structures & Algorithms, OOPS, OS, Computer Architecture, Computer Networks, ML Compilers, Distributed systems)

EXPERIENCE

Synaptics Incorporated - Apprentice | USA [LINK] — ***March 2025- Dec 2025***

(Embedded & IOT, ARM + NPU, CV, ML, Yocto Linux)

- Built an **embedded navigation assistant** for *visually impaired users*, led embedded Linux bring-up on **Synaptics Astra SL1680** with **Yocto**, headless deployment, multithreaded C++/Python services; stabilized inference pipelines to achieve **<100 ms latency at ~1 W**.
- Brought up and validated camera pipelines using **Sony IMX477** patching **V4L2** nodes and tuning exposure, gain, and color; collected and curated **55k+ images** and trained detection models with **Roboflow**, to improve **real-time object detection** robustness.
- Deployed optimized **YOLOv8-n** models (**ONNX → TFLite → Synap.rt**) using Docker, implementing NEON-accelerated pre/post-processing, *temporal gating*, and *decision smoothing* to reduce false "Walk / Don't-Walk" events by **~38%** in live operation.
- Performed low-level **GPIO** and **hardware bring-up**, validating GPIO and peripheral interfaces (**I^2C, SPI**) with scope-based verification; integrated **ultrasonic** sensing for **obstacle avoidance** & **IMU-based** fall detection and implemented, **Twilio-based SOS** alert system.

Embedded System Intern at Envipco | USA *(CV Integration, STM32, Edge ML)* — ***Jul 2025 – Dec 2025***

- Built on-device computer vision for RVM *(Reverse Vending Machines)* using **OpenMV AE3** with **YOLOv8-style TFLite** inference and **NMS**; serialized detections (bottle/hand/logo + barcode ROI) over **UART/CAN** to drive the main controller **FSM**.
- Led **360° multi-camera barcode ring** evaluation (5 to 7 imagers), testing scanner/optics/firmware settings; built Python tooling (multi-serial orchestration + logging) to measure coverage gaps and **first-decode latency**, enabling sensor selection.
- Delivered production **STM32 (ARM Cortex-M)** firmware: **TIM-triggered ADC + DMA ISR/RTOS** pipelines with calibration and fixed-point features; re-architected the material-ID coil ring (**7 to 4 sensors**) and integrated **RS-485/Modbus + CAN**; brought up a **LiDAR bin-fill controller** (Altium PCB) with SPI/I^2C/ADC, **CRC-16**, and bootloader.

System Engineer at Atlas Copco | India *(Embedded Firmware, RFID, RTOS, Sensor Interfaces)* — ***Jan 2024 – June 2024***

- Implemented **RFID PHY/MAC firmware** in C/C++, with frame encoding/CRC, anti-collision logic, RF front-end calibration (modulation, AGC, RSSI), and **packetization + encryption** at the link layer to improve read stability and data integrity.
- Built **FreeRTOS-based firmware** for RFID readers, integrating SPI/I^2C/UART drivers with ISR/DMA paths and deterministic timing for a real-time embedded system; handling data packet transfer **at PHY/MAC layers** of the stack.
- Optimized antenna and EMI behavior; ran read-range and detuning tests, and documented TX/RX verification, **frame/token handling**, and factory bring-up procedures for repeatable hardware integration.
- Set up **Git CI/CD pipelines** with unit, integration, & HIL tests, cutting release cycles by ~40% and improving firmware reliability; participated in **code reviews** to uphold coding standards, security considerations (encryption/token use), and maintainability.

PROJECTS

Light Swarm-IoT-System *(Embedded IoT, Distributed Systems, Raspberry Pi, SPI)* **[LINK]** — ***Sept 2024 – Dec 2025***

- Built a distributed swarm of **5 ESP8266 nodes + Raspberry Pi** with dynamic master election, using **UDP multicast** and a structured packet protocol; added smoothing and collision-aware broadcast logic for stable real-time operation.
- Implemented the Pi hub in Python for packet parsing, logging, and master-transition tracking; drove a **MAX7219 8×8** SPI LED matrix for live visualization and integrated **Node-RED/Grafana** dashboards for latency and *packet-loss KPIs.*

Real-Time Canny Edge Detector *(C/C++ & DSP - Linux PREEMPT-RT)* **[LINK]** — ***Jan 2025 – March 2025***

- Built a real-time C/C++ Canny pipeline on **PREEMPT-RT**, leveraging **GPU** offloading for edge-detection acceleration, sustaining **30 FPS @ 1080p** within a fixed latency budget using **ARM NEON SIMD**, cache-aware tiling, and **POSIX threads**.
- Reached **98%** edge accuracy with **CPU <70%**; profiled with **gprof**, parallelized blur stages, and modeled worst-case timing to ensure **deterministic** execution.

SKILLS

Programming:	C, C++, Python
Embedded & RTOS:	ARM Cortex-M, STM32, FreeRTOS, ISR/DMA, concurrency, memory mgmt.
Interfaces/Networking:	SPI, I^2C, UART, CAN, Ethernet, RS-485 (Modbus), TCP/UDP, BLE, Wi-Fi/LTE
Operating Systems:	Linux (Yocto, PREEMPT-RT), POSIX, Windows, RTOS, Ubuntu
Debug & Test:	Oscilloscopes, GDB, Wireshark logic analyzers, HIL, unit/regression (Unity/C, pytest)
Tooling:	Git, Postman, CI/CD (GitLab/Jenkins), Docker, Altium

LEADERSHIP & STARTUP EXPERIENCE

Co-Founder – Transfusiotrack (Blood Bag Delivery System) *(Full Stack | Cloud | CI/CD)* **[LINK]** — ***June 2023 – Present***

Co-founded a healthcare logistics platform to reduce blood delivery delays; led product and engineering with **Flask**, **MongoDB**, and **REST APIs**, onboarded blood banks/hospitals, and integrated payment gateways for 100% cashless payments (UPI/cards).

Deployed and scaled on **AWS (Lambda, EC2)** before migrating to **Render;** built **Docker + Git CI/CD**, cutting releases by **~40%** and improving delivery coordination by **~30%** across two cities; handled stakeholder coordination and operational reporting.